BTS-VI(SS)-01.19-0012 Reg. No.

A

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1602 MACHINE DESIGN I

(2015 Scheme)

(Use of approved design data book is permitted. Assume suitable data, if necessary)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) What are the factors affecting the selection of materials?

(b) What is stress concentration? Explain the methods to reduce its effect.

(c) What is meant by creep? Discuss the design considerations.

(d) Why is an angular thread preferred in fastening?

(e) What is the special advantage of a flexible coupling over a rigid type?

(f) Briefly explain the term efficiency of a riveted joint.

(g) What is concentric helical spring and where does it find applications.

(h) What are the advantages and disadvantages of welded joints?

(i) Explain a method to obtain a bolt of uniform strength.

(j) What is meant by critical speed of shafts?

PART B

$(4 \times 10 = 40)$

II. (a) Distinguish between static and impact loads. (2)

(b) Determine the diameter of a steel shaft using any two theories of failure subjected to a bending moment of 12 kNm and a twisting moment of 32 kNm. Assume yield strength as 800 N/mm^2, factor of safety as 3. (8)

OR

III. Determine the diameter of a circular rod made of ductile steel having ultimate tensile strength of 600 N/mm^2 and tensile yield strength of 300 N/mm^2. The member is subjected to a varying bending moment from 200 Nm clockwise and 500 Nm anticlockwise and a twisting moment varying from 50 Nm to 150 Nm. (10)

Stress concentration factor = 1.8, Factor of safety = 2

IV. A double threaded power screw is used to raise a load of 6 kN. The nominal diameter is 60 mm and the pitch is 9 mm. The threads are acme type and the coefficient of friction at the screw thread is 0.15. Neglecting collar friction, find the torque required to raise and lower the load and also efficiency of the screw for lifting load. (10)

OR

(P.T.O.)

V. Design and draw a knuckle joint made of plain carbon steel with an axial load of 90 kN. Tensile yield strength = 90 N/mm^2, shear stress = 60 N/mm^2 and crushing stress = 150 N/mm^2. (10)

VI. Design a double riveted butt joint with two cover plates for the longitudinal seam of a boiler shell 1.5 m in diameter subjected to a steam pressure of 0.9 N/mm^2. Assume joint efficiency as 75%, allowable tensile stress in the plate 90 N/mm^2, and compressive stress 150 N/mm^2 and shear stress in the rivet 60 N/mm^2. (10)

OR

VII. A locomotive semi elliptical laminated spring has an overall length of 1 m and sustains a load of 70 kN. The spring has 3 full length leaves and 15 graduated leaves with a central band of 100 mm. All the leaves are to be stressed to 400 N/mm^2 when fully loaded. The ratio of spring depth to width is 2. Determine the thickness and width of the leaves. Also find the initial gap that should be provided between the full and graduated leaves before band load is applied. (10)

VIII. (a) A 50 mm diameter solid shaft of length 200 mm is welded to a flat plate by a fillet weld along the circumference of the shaft. The shaft is subjected to a vertical load of 10 kN at its free end. Determine the size of the weld if the allowable shear stress in the weld is 100 N/mm^2. (5)

(b) A 200 × 100 × 10 mm 'angle' is welded to a frame at top and bottom. The 'angle' is subjected to a static load of 150 kN axially through the centroid of the angle plate. Determine the lengths of the weld at the top and bottom. Assume the permissible shear stress = 70 N/mm^2. (5)

OR

IX. A hollow shaft of 400 mm outside diameter and 250 mm inside diameter is supported on two bearings 3 m apart. The shaft rotates at 200 rpm and transmits 130 kW of power. The weight of the shaft is 90 kN and it receives a thrust load of 300 kN. Determine the maximum normal and shear stress induced. (10)

***